

Anthony Ellis

Software Engineer — Simulation, Graphics & Tooling

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SUMMARY

Self-taught engineer who builds real-time simulations from scratch in JavaScript — procedural generation, agent-based traffic, flocking, and the profiling work that holds 60 fps with no engine underneath. Currently supporting driverless vehicle operations at Zoox: resolving off-nominal situations remotely, testing pre-release features and filing the reports engineering works from. Seeking simulation, gameplay, graphics or tooling work.

PROJECTS

Procedural Galaxy Generator · [three.js](#), [GLSL](#), [Web Workers](#)

anthonyellis.dev/projects/galaxy

- Renders **250,000 stars in a single draw call at 60 fps** — 14 bytes per star, with sub-pixel dimming in a custom shader so distant galaxies attenuate correctly instead of getting brighter.
- Every galaxy, star, planet and terrain patch is a **pure function of one 48-bit seed** — the URL is the save file, any bug reproduces exactly, and the LOD cache can evict freely because nothing is ever the only copy.
- Headless-Chromium suite asserts **bit-identical** star buffers across runs and reads back the framebuffer per view, so a silently black render fails the build.

Arterial — City Traffic Simulator · [Canvas 2D](#), [graph algorithms](#)

anthonyellis.dev/projects/arterial

- Agent-based traffic where congestion is emergent, not scripted: **Intelligent Driver Model** car-following, **MOBIL** lane changing, and **A*** routing over lane space with turn, jam and lane-change penalties.
- Junctions arbitrate by reservation over a pre-computed conflict set; fixed 60 Hz sim with interpolated rendering, and roundabouts implemented as a **graph rewrite** so the vehicle physics needs no special cases.

Aquarium & Cellular Automata Sandbox · [Canvas 2D](#), [performance](#)

anthonyellis.dev/projects/aquarium

- Scales flocking from hundreds to **4,000 agents** by replacing $O(n^2)$ neighbour search with a spatial hash over structure-of-arrays typed buffers — **zero allocation per frame**.
- Agents steer around a falling-sand cellular automaton the user reshapes in real time using three $O(1)$ grid probes each, and the renderer degrades by dropping detail at population thresholds rather than dropping frames.

EXPERIENCE

Ride Operations Specialist — Zoox (contract)

Jan 2026 – Present

Las Vegas, NV · autonomous vehicle ride-hailing

- Monitor and remotely operate driverless vehicles in live public service — camera surveys, door control, two-way rider audio — resolving off-nominal situations in real time under time pressure.
- Run **structured test missions on pre-release features**, file defect reports with the reproduction detail engineering needs, and review and label data used for **autonomy model training**.
- Wrote and maintained onboarding material now used to train new operators; coordinate across engineering, support, venue security and law enforcement during incidents.

EARLIER EXPERIENCE

Customer Service Representative — Fasteners, Inc. / Southwestern Supply, Las Vegas, NV · Jul 2023 – Dec 2024

Technical product support and order management in **SAP** and **NetSuite**; matched customers to the correct parts from spec.

Customer Service Representative — Alorica, Las Vegas, NV · Jan 2019 – Jun 2023

High-volume call and email support; resolved escalated complaints by tracing root cause, consistently beating resolution targets.

Solar Appointment Setter — SunSource Energy, Las Vegas, NV · Jan – Jul 2025

Cold-call lead generation, 3–5 qualified appointments per day against target; company has since closed.

TECHNICAL SKILLS

Languages	JavaScript (ES2022), GLSL, HTML, CSS
Graphics & sim	WebGL, three.js, Canvas 2D, custom shaders, level-of-detail systems, procedural generation, deterministic PRNG & seed chains, noise, cellular automata, boids/steering, IDM & MOBIL traffic models, fixed-timestep loops
Algorithms	A* and graph search, spatial hashing, structure-of-arrays layouts, typed arrays, object pooling, frame-budget profiling
Tooling	Node.js, Web Workers, Playwright / headless Chromium testing, Chrome DevTools, Git, GitHub Pages

EDUCATION

Del Sol High School — High school diploma, Las Vegas, NV

2014